

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Technical Presentation

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO1 – Imitation (L1)	Assessment:	Assessment Tool 2 – Technical Presentation
Weightage:	30% of CIA		
CO1 Statement:	Analyse NLP models, regular expression patterns, finite state automata, morphological processing techniques and to interpret language processing. (BL-3)		
Student Name:	HARIHARAN G	Reg. No.:	192424156
Section / Batch:	B.Tech AI/DS	Date:	16/07/2026

Code of Conduct

I HARIHARAN G 192424156(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: HARIHARAN G

Instructions

- Suggested Team Member Roles (3 Members)
- Member 1: Theory, Concepts, and Literature Review
- Member 2: Algorithm Design, Models, and Technical Analysis
- Member 3: Demonstration, Case Study, Applications, and Conclusion
- GPS photo must submit.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
Evolution of Natural Language Processing: From Rule-Based Systems to Modern NLP	NLP Models, Language Processing Evolution	TP-01
Comparative Study of NLP Models and Algorithms	Model Analysis, Algorithm Comparison	TP-02
Regular Expressions for Real-World Text Processing	Pattern Matching, Information Extraction	TP-03
Designing Efficient Regular Expression Patterns for Information Extraction	Text Mining, Data Extraction Techniques	TP-04
Finite State Automata in Natural Language Processing	State Transitions, Language Recognition	TP-05
Building a Finite State Automaton for Token Recognition	Lexical Analysis, Token Validation	TP-06
Morphology and Word Formation in Natural Languages	Morpheme Analysis, Word Structure Processing	TP-07
Inflectional Morphology: Understanding Grammatical Variations	Grammatical Analysis, Word Normalization	TP-08
Derivational Morphology and Vocabulary Expansion	Word Generation, Semantic Transformation	TP-09
Comparative Analysis of Inflectional and Derivational Morphology	Morphological Classification, Language Analysis	TP-10

Finite-State Morphological Parsing Techniques	Morphological Parsing, Finite-State Processing	TP-11
Morphological Analysis for Indian Languages	Language-Specific Morphology, NLP Challenges	TP-12
Porter Stemmer: Design, Working Mechanism, and Applications	Stemming Algorithms, Text Preprocessing	TP-13
Stemming vs Morphological Parsing: A Comparative Study	Word Reduction, Morphological Processing	TP-14
Integrated NLP Pipeline: From Regular Expressions to Morphological Processing	End-to-End Language Processing Pipeline	TP-15

Weighted Finite-State Transducers (WFSTs)

The Hidden Engine Behind Modern Natural Language Processing

Presented By

Hariharan G

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Artificial Intelligence And Data Science

Introduction

- Natural Language Processing (NLP) enables computers to understand and process human language.
- Human language is often ambiguous, producing multiple possible interpretations.
- Traditional Finite State Automata (FSA) can only recognize patterns.
- WFST extends FSA by generating outputs and assigning probabilities (weights).
- It selects the most probable output automatically.

Why Traditional FSA is Not Enough

Limitations of Finite State Automata

- Accepts or rejects input strings only.
- Cannot transform one word into another.
- Does not rank multiple outputs.
- Cannot represent probabilities or confidence.
- Fails to resolve ambiguity in speech and language processing.

Example

Input:

Ice cream

Possible interpretations:

- Ice cream
- I scream

FSA cannot determine which one is correct.

From FSA to WFST

Evolution of Finite-State Models



Model	Output	Weights
FSA	✗	✗
FST	✓	✗
WFST	✓	✓

What is a WFST?

A **Weighted Finite-State Transducer (WFST)** is a mathematical model that:

- Reads an input sequence.
- Produces an output sequence.
- Assigns a numerical weight to every transition.
- Chooses the path with the lowest total cost or highest probability.
- Combines language rules with probability.

Components of WFST

Main Components

- Every transition contains:
- Current State
- Input Symbol
- Output Symbol
- Weight (Cost / Probability)
- Next State

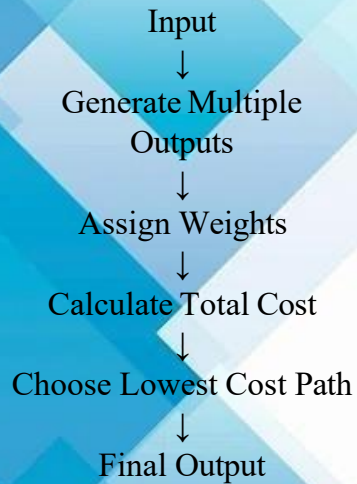
Example

q0 cat : CAT / 0.2 ↓ q1

Lower weight = Higher probability

How WFST Works

Working Process



Example

Output	Cost
They're	0.2
There	0.8
Their	1.4

Result → **They're**

Mathematical Representation

Formal Representation

- A WFST is represented as:

$$T = (Q, \Sigma, \Gamma, E, I, F)$$

Where:

- **Q** – Set of States
- **Z** – Input Alphabet
- **Γ** – Output Alphabet
- **E** – Weighted Transitions
- **I** – Initial State
- **F** – Final States

Transition:

(Current State,
Input,
Output,
Weight,
Next State)

Algorithms Used

- **Path Selection Algorithms**
- WFST searches for the best path using:
- **Viterbi Algorithm**
 - Finds the most probable sequence.
- **Dijkstra Algorithm**
 - Finds the minimum-cost path.
- **Shortest Path Search**
 - Improves decoding speed and accuracy.

Real-Time Applications

Applications of WFST

- Speech Recognition
- Spell Correction
- Machine Translation
- Morphological Analysis
- OCR (Optical Character Recognition)
- Search Engines
- Text Normalization
- Voice Assistants (Siri, Alexa)

Advantages

- Very fast processing.
- Low memory consumption.
- Handles ambiguity efficiently.
- Mathematically explainable.
- Easy to optimize.
- Suitable for embedded systems.
- High accuracy in rule-based NLP tasks.

Limitations

- Cannot understand semantic meaning.
- Requires manually designed rules.
- Large WFST graphs become complex.
- Poor at handling long-distance context.
- Less adaptive than deep learning models.

WFST vs Neural Networks

Feature	WFST	Neural Networks
Speed	High	Medium
Memory	Low	High
Explainability	Excellent	Low
Training Required	No	Yes
Context Understanding	Limited	Excellent
Adaptability	Rule-Based	Learns from Data

Future Scope

Future of WFST

- Hybrid AI + WFST systems.
- Efficient on-device speech recognition.
- NLP for low-resource languages.
- Embedded AI applications.
- Text normalization for conversational AI.
- Integration with Transformer-based models.

Conclusion

- WFST extends Finite State Transducers by introducing weighted transitions.
- Weights represent probabilities or costs.
- The lowest-cost path produces the most likely output.
- Widely used in speech recognition, spell correction, translation, and morphology.
- WFST remains an important technology alongside modern deep learning approaches.



Annexure A – Technical Presentation

Team No.	Technical Presentation Title	Description
1	Evolution of Natural Language Processing: From Rule-Based Systems to Modern NLP	Analyze the historical development of NLP models, key milestones, and how language processing has evolved from handcrafted rules to intelligent systems.
2	Comparative Study of NLP Models and Algorithms	Explore different NLP models and algorithms used for language processing, comparing their working principles, strengths, and limitations.
3	Regular Expressions for Real-World Text Processing	Demonstrate how regular expressions are used for data validation, text extraction, log analysis, and information retrieval in practical applications.
4	Designing Efficient Regular Expression Patterns for Information Extraction	Analyze complex regular expression patterns used to extract emails, phone numbers, URLs, dates, and structured information from unstructured text.
5	Finite State Automata in Natural Language Processing	Explain the concepts of deterministic and non-deterministic finite automata and their role in lexical analysis and language recognition.
6	Building a Finite State Automaton for Token Recognition	Design and demonstrate a finite state automaton that recognizes tokens such as identifiers, keywords, and numbers in textual data.
7	Morphology and Word Formation in Natural Languages	Investigate how words are structured and formed using morphemes, and analyze the importance of morphology in NLP systems.
8	Inflectional Morphology: Understanding Grammatical Variations	Examine how inflectional changes affect word forms based on tense, number, gender, and case across different languages.
9	Derivational Morphology and Vocabulary Expansion	Analyze derivational processes that create new words and study their impact on semantic interpretation and language understanding.
10	Comparative Analysis of Inflectional and Derivational Morphology	Compare the characteristics, applications, and challenges of inflectional and derivational morphology in NLP tasks.
11	Finite-State Morphological Parsing Techniques	Explore finite-state transducers and parsing techniques used to analyze and generate morphological structures in natural languages.
12	Morphological Analysis for Indian Languages	Study the challenges and solutions involved in performing morphological processing for morphologically rich Indian languages such as Tamil, Hindi, and Telugu.
13	Porter Stemmer: Design, Working Mechanism, and Applications	Analyze the Porter Stemming algorithm, its rule-based approach, implementation steps, and role in information retrieval systems.
14	Stemming vs Morphological Parsing: A Comparative Study	Compare stemming and morphological parsing techniques based on accuracy, computational complexity, and application scenarios.
15	Integrated NLP Pipeline: From Regular Expressions to Morphological Processing	Demonstrate a complete NLP preprocessing pipeline incorporating regular expressions, finite state automata, stemming, and morphological analysis for text processing.

Rubrics for Calculation

Technical Presentation					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Team No. / Criteria Level	Technical Content Accuracy	Problem Identification	Use of Tools	Communication	Professionalism	Total %
1.	18	19	18	18	19	92

Faculty In-charge	Course Coordinator	Program Director
Dr.T.Kumaragurubaran		
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____